Hetauda School of Management

Department of Information Technology

First Terminal Examination:2024

Course Title: Digital Logic **Full marks: 60**
Code No: CACS101 **Pass marks: 24**
Semester: 1 **Time: 3 hrs.**

Candidates are required to give their own words as far as Practicable.

Attempt any Six questions. **Group B** **[6*5 =30]**

1. . Define analog and digital signals. What are the advantages of digital signals over analog signals?
2. State and prove commutative, distributive law, and De Morgan's theorem using the truth table method.
3. Perform the following number system conversions.
   a) $(1101.101)_2 = (?)_{10}$
   b) $(27.625)_{10} = (?)_5$
   c) $(101101)_2$ to octal and Hexadecimal number system.
4. How are negative numbers represented in binary form? Explain with an example.
5. Subtract $(1010.110)_2 - (101.101)_2$ using both 1's complement and 2's complement methods.
6. Explain basic logic gates and universal gates. Why is NOR gate a universal gate?
7. Explain r's complement and (r-1)'s complement with suitable examples.

Attempt any Two questions **Group C** **[10 × 2 = 20 Marks]**

8. What is the difference between canonical form and standard form? Which form is preferable when implementing a Boolean function with gates? Which form is obtained when reading a function from a truth table?
9. Express the given Boolean function in Sum of Minters (SOM) and Product of Maxterm's (POM):

$$f(A, B, C, D) = (A+B' +C)\ (A+B')\ (A+C'+D')\ (A'+B+C+D')\ (B+C'+D')$$

10. Differentiate between Sum of Products (SOP) and Product of Sums (POS) with a suitable example. Express the given Boolean function in Sum of Minterms (SOM) and Product of Maxterms (POM): $f(A,B,C,D)=D(A'+B)+B'D$.
11. Write short notes on the following:
    a) Excess-3 (XS-3) Code
    b) Binary Codes
    c) Error Detection Codes
    d) Derived Gates